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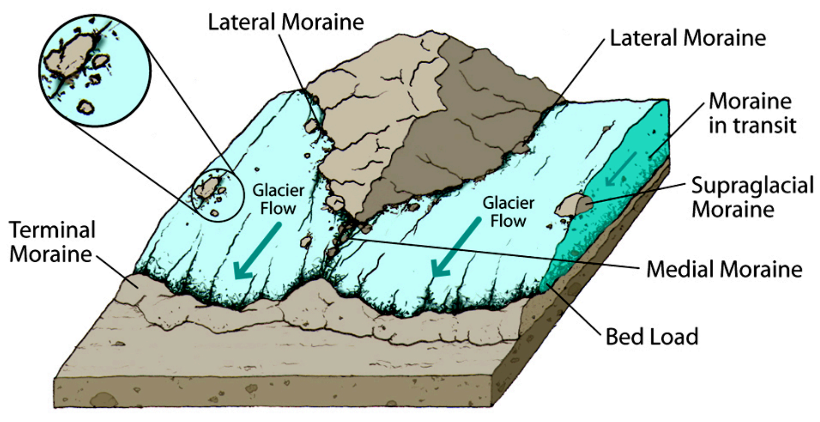
Reconstruction of the Glacier Fronts Trajectories by Stochastic Simulations

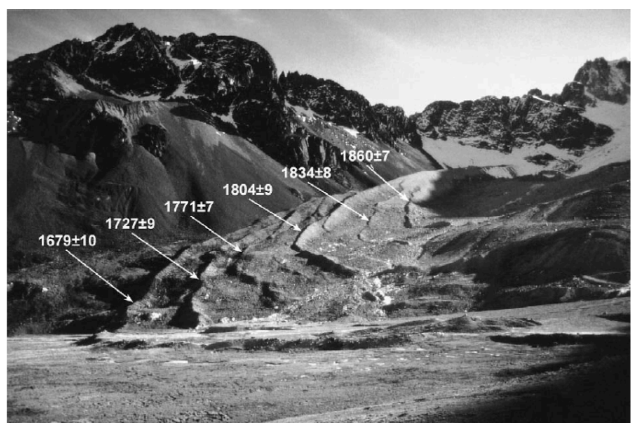
Mer de Glace (France)

Context : Moraines, Evidence of Glacial Changes

The retreat of glaciers is due in part to climate conditions at high elevation sites and may create instability (ice falls, avalanches, etc.). Visible **moraines** are evidence of past glacier positions.
They correspond to the **records** R of the series of the **local maxima** of the front relative positions when the time axis is reversed. $\{R_n\}_{n\geq 1} = \{X_t > \max(X_1, ..., X_{t-1}, X_{t+1})), t \geq 1\}$.

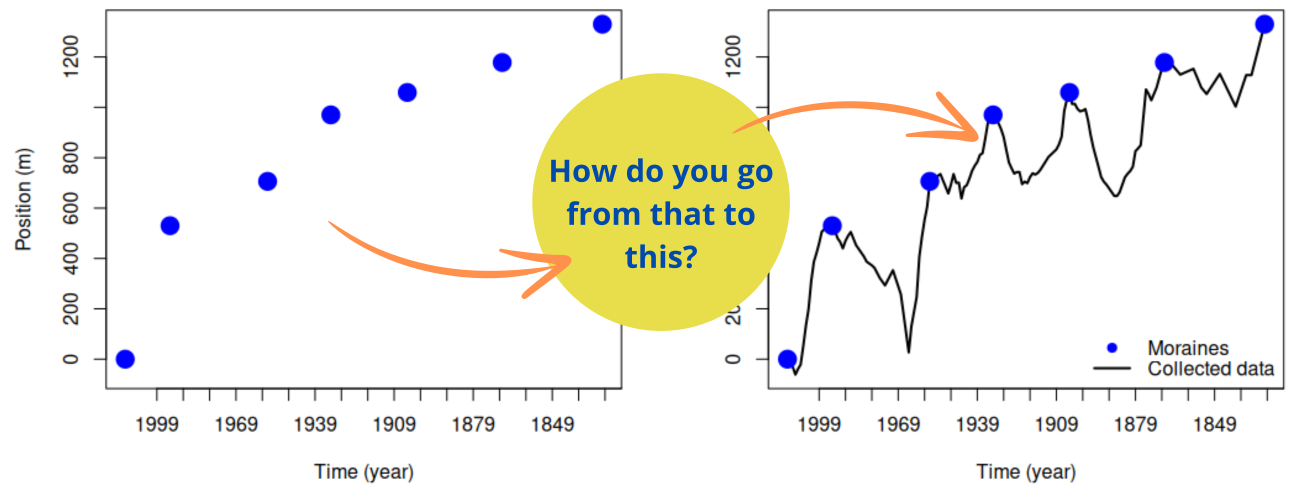


Moraine formation schema



Moraines of the Charquini glacier, Bolivia

Moraines and front variation of the Glacier *des Bossons*



Objective : Build a probabilistic model tracing past glacier front variation trajectories using their moraines

Goal : Apply it to existing moraine series and propose physically plausible trajectories
Challenges : Respect physical constraints

Model : How to simulate trajectories from given records ?

From $(B_t)_{t\in[0,T]}$ a brownian motion to $(X_t)_{t\in\{1,...,\mathcal{T}\}}$ a valid trajectory between the moraines R_1 and R_2.

1) Brownian motion simulation :
A continuous-time stochastic process $(B_t)_{t\in[0,T]}$ (with T big enough) is a brownian motion if it :
(i) Is almost-surely continuous in time
(ii)Respects $B_0 = 0$
(iii) Has zero-mean Gaussian independent increments : $B_t - B_s \sim \mathcal{N}(0, \sigma^2(t-s))$ for any $0 \leq s < t$, and $B_t - B_s$ is independent of $B_{t'} - B_{s'}$ whenever $0 \leq s < t \leq s' < t'$

Variance hyper parameter

2) Brownian bridge construction :

$$\tilde{B}_t = B_t - \frac{t}{T}B_T, \quad t \in [0, T]. \tag{1}$$

3) Configuration of start and end points :

$$\tilde{B}_t^{x,y} = x + \frac{t}{T}(y - x) + \tilde{B}_t, \quad t \in [0, T]. \tag{2}$$

4) Discretisation and application to moraines R_1 and R_2:

$$Y_t = \tilde{B}_{t-1}^{R_1,R_2}, \quad t \in \{1, ..., \mathcal{T}\}. \tag{3}$$

5) Definition of the Vervaat transform V that maps a time series $\{F_t\}_{t\in\{1,...,\mathcal{T}\}}$ to a time series $\{V(F)_t\}_{t\in\{1,...,\mathcal{T}\}}$:

$$t \mapsto V(F)_t = \begin{cases} F_1 - F_{T_{max}} + F_{T_{max}-1+t}, & \text{if } 1 \leq t \leq 1 + \mathcal{T} - T_{max}, \\ F_{\mathcal{T}} - F_{T_{max}} + F_{T_{max}-\mathcal{T}+t}, & \text{if } 1 + \mathcal{T} - T_{max} < t \leq \mathcal{T}, \end{cases} \tag{4}$$

where

$$T_{max} = \min\left\{\underset{t\in\{1,...,\mathcal{T}\}}{\operatorname{argmax}} F_t\right\}, \tag{5}$$

$$Z_t = V(Y)_t, \quad t \in \{1, \ldots, \mathcal{T}\} \tag{6}$$

6) Smoothing:

$$X_t \approx \tilde{Z}_t \tag{7}$$

(Gaussian smoothing of $\{Z_t\}_{t\in\{1,...,\mathcal{T}\}}$ with bandwidth s)

Smoothing hyper parameter

PEPR Risques (IRiMa)

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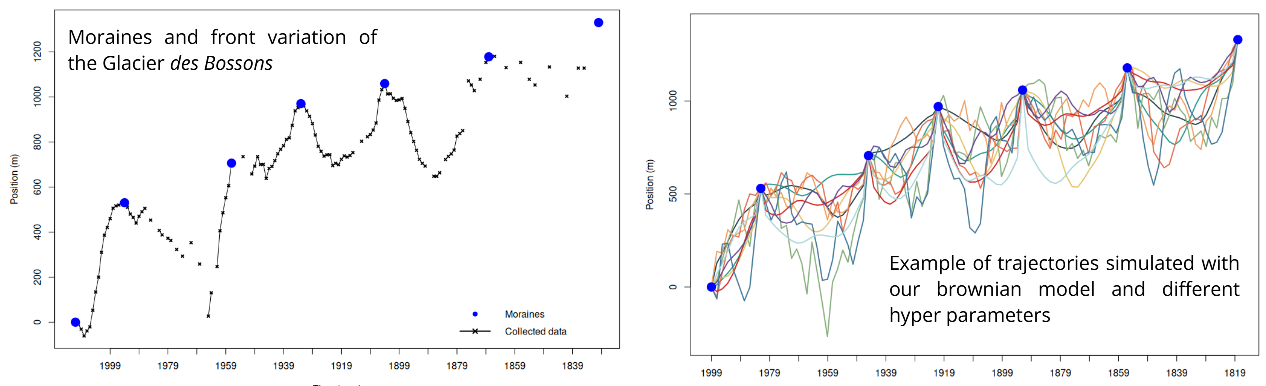
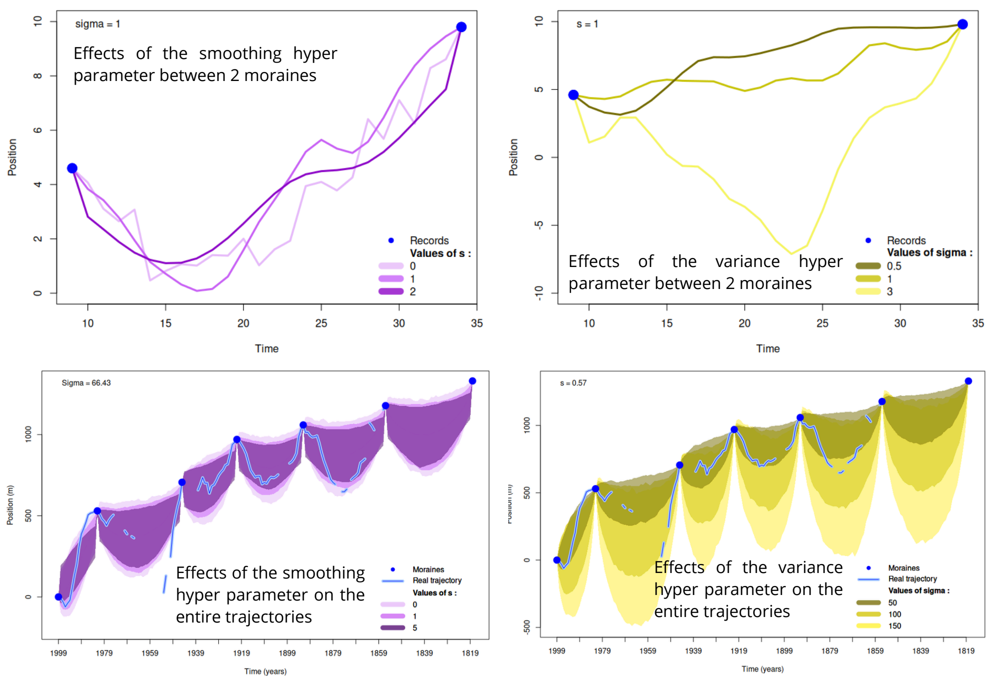
Application to the Glacier *Des Bossons* (Alps, France)

Data : an almost complete trajectory measured for more than 150 years
What can our model do so far : Simulate some very different front variation trajectories, depending on the **hyper parameters** values

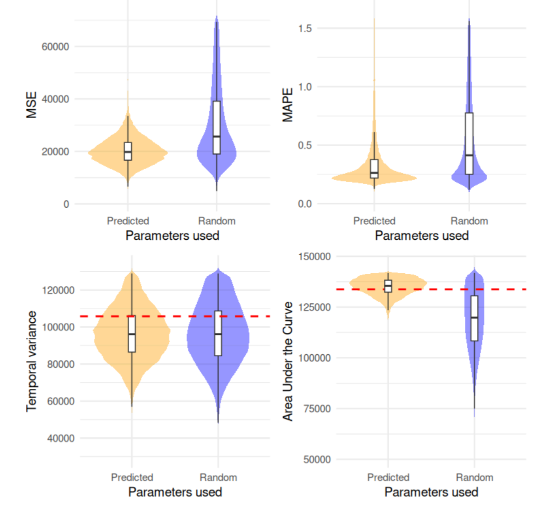


Calibration : Effects of the hyper parameters (hp) on the trajectories



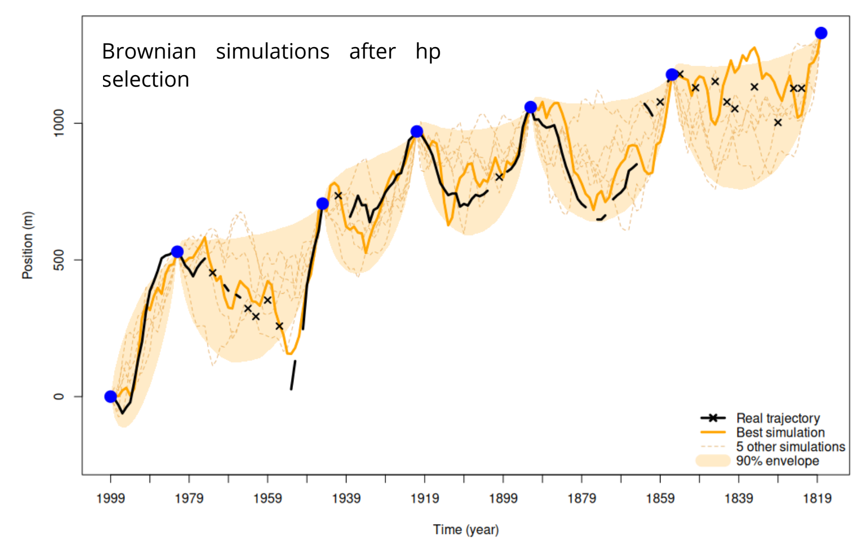
To chose the **best set of hyper parameters** we cannot compute the likelihood of our model because of the smoothing.
Instead we can simulate a lot of trajectories with different hp and train a **Neural Network** model to learn the link between those hp and the simulated trajectories. Then we can select the hp that allow our model to simulate trajectories that are close to the real one.

Results : Effects of the hp selection



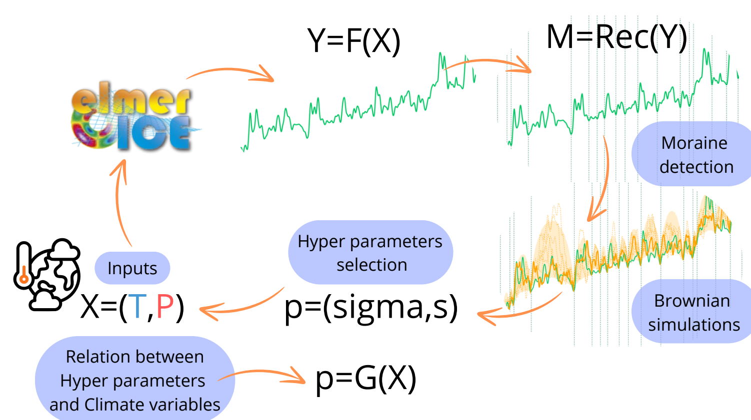
Metrics comparision between the real trajectory and simulations made respectively with and without hp selection

When we give the real trajectory (or a part of it) to our Neural Network model we obtain an hp set which allows us to simulate trajectories really close to the reality.



Work in progress

- Use **Elmer Ice** front variation simulations that have been simulated with some climate variables (P,T) in order to use our brownian model on it and find the best hp set. Link those hp with the climate input variables to be able to chose the hp when we have no data available except the moraines.
- Extend the model to a **multivariate** setting to take into account the correlation between front variation trajectories of glaciers that are located close from each other.



Stochastic generator of trajectories from record data: application to the fluctuations of a glacier's frontal position from a sample of moraines : Paper submitted in Environmetrics this year

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